

VERIFIED AND EXECUTABLE LINEAR ALGEBRA

DenseMatrix: a computationally efficient matrix representation in Lean.

Row-major array representation for matrix computations, with a ring isomorphism to the proof-friendly mathlib Matrix.

TEAM

Annis Wu · Joseph Qian · Junye Ji
· Veer Shukla

Mentor: Dhruv Bhatia



SCAN FOR
ARTIFACT

Why DenseMatrix

- **The Problem:** Current mathlib representation is function-backed. This is great for theory, but terrible for computation speeds.
- **The Solution:** We defined a new matrix representation using a row-major 1D array to improve runtime for linear algebra algorithms.
- **The Bridge:** We proved a ring isomorphism exists between DenseMatrix and mathlib Matrix. Algorithms are written in DenseMatrix for speed, but proven using mathlib Matrix (transporting theory via the isomorphism).
- **Next Steps:** Porting our existing algorithms to this new, computationally-efficient data type.

Tail Recursion

Lean uses a functional but in-place paradigm: it optimizes memory by updating values in-place *only* if there is a single reference to that value. If the reference count exceeds 1, array mutation requires $O(n)$ time, significantly slowing down algorithms.

While the caller ensures the initial matrix has a reference count of 1, we rely on *tail recursion* to ensure that count doesn't increase during recursive multiplication.

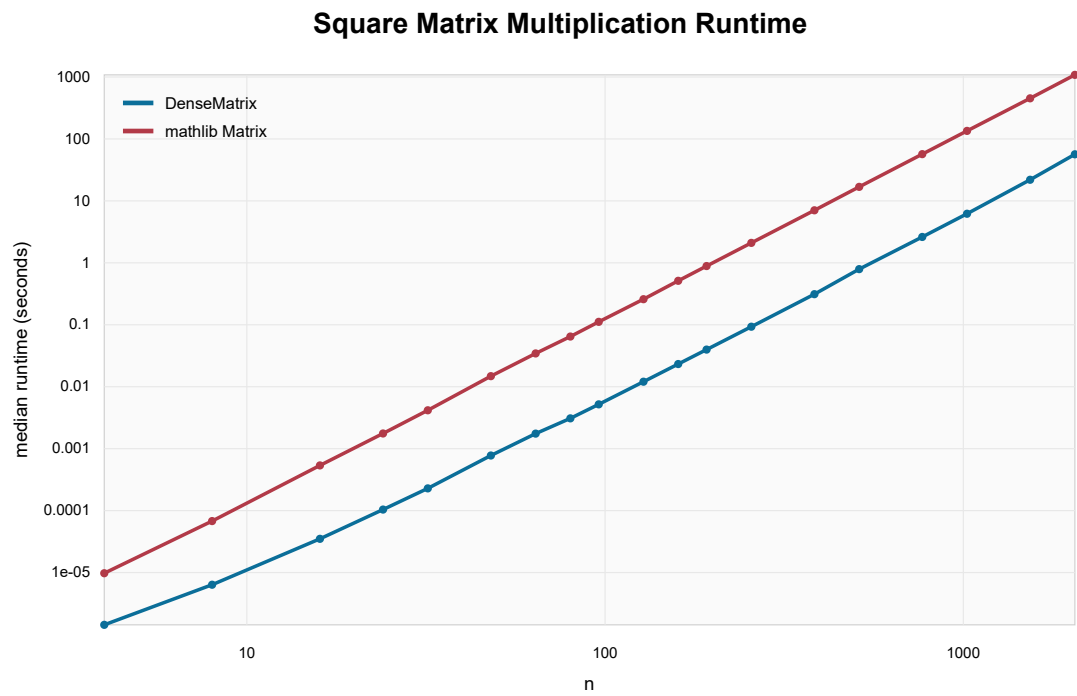
Functional languages like Lean use "tail-call elimination". If a function call is in "tail position," the compiler replaces the current stack frame instead of pushing a new one. This keeps the reference count constant and ensures efficient array mutation.

Definition: A function call is in tail position if the caller doesn't need to modify its returned value before returning. Notably:

- If a match-expression is in tail position, each branch is too.
- If an if-expression is in tail position, both branches are too.
- If a let-expression is in tail position, its body is too.

Multiplication Scaling

DenseMatrix vs mathlib Matrix multiplication median runtime



Representation Tradeoff

DenseMatrix vs mathlib Matrix median speedup

